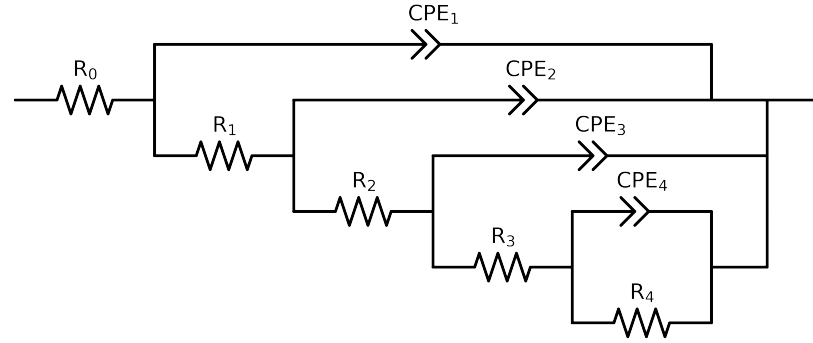


Comparative EIS Kinetics Report

Model Structure: R0-p(R1-p(R2-p(R3-p(R4,CPE4),CPE3),CPE2),CPE1)

Used data: original data



Element	Unit	Bounds	Vasco_LNV3_4	Vasco_LNV3_5	Vasco_LNV3_7d_
R_0	Ω	$[1.0e+00, 1.0e+03]$	$1.84e+02 \pm 5.2e+00$	$1.25e+02 \pm 2.6e+00$	$1.15e+02 \pm 1.3e+00$
R_1	Ω	$[1.0e+00, 1.0e+08]$	$4.90e+05 \pm 1.7e+05$	$8.28e+02 \pm 7.4e+01$	$4.69e+02 \pm 4.4e+01$
R_2	Ω	$[1.0e+00, 1.0e+08]$	$1.11e+03 \pm 3.2e+10$	$4.19e+05 \pm 9.9e+04$	$1.46e+03 \pm 2.2e+02$
R_3	Ω	$[1.0e+00, 1.0e+08]$	$5.88e+05 \pm 6.1e+11$	$1.02e+00 \pm 2.9e+06$	$2.43e+05 \pm 8.6e+03$
R_4	Ω	$[1.0e+00, 1.0e+08]$	$1.73e+04 \pm 6.1e+11$	$8.17e+06 \pm 1.0e+14$	$6.51e+05 \pm 7.0e+04$
Q_4	$\Omega^{-1} \text{ sec}^a$	$[1.0e-09, 1.0e-03]$	$1.27e-09 \pm 1.4e-01$	$1.54e-06 \pm 1.6e+00$	$4.15e-05 \pm 3.5e-06$
n_4	-	$[0.0e+00, 1.0e+00]$	$6.26e-05 \pm 2.9e+06$	$1.10e-11 \pm 1.4e+00$	$9.43e-01 \pm 3.3e-02$
Q_3	$\Omega^{-1} \text{ sec}^a$	$[1.0e-09, 1.0e-03]$	$1.00e-09 \pm 7.7e-02$	$4.54e-05 \pm 5.0e-05$	$5.80e-06 \pm 1.0e-06$
n_3	-	$[0.0e+00, 1.0e+00]$	$1.00e+00 \pm 3.3e+06$	$1.00e+00 \pm 7.1e-01$	$8.76e-01 \pm 2.1e-02$
Q_2	$\Omega^{-1} \text{ sec}^a$	$[1.0e-09, 1.0e-03]$	$3.63e-05 \pm 7.6e-02$	$5.96e-06 \pm 8.7e-07$	$3.80e-06 \pm 1.5e-06$
n_2	-	$[0.0e+00, 1.0e+00]$	$1.00e+00 \pm 9.1e+01$	$8.75e-01 \pm 2.1e-02$	$8.27e-01 \pm 5.1e-02$
Q_1	$\Omega^{-1} \text{ sec}^a$	$[1.0e-09, 1.0e-03]$	$9.02e-06 \pm 4.4e-07$	$5.85e-06 \pm 1.2e-06$	$3.29e-06 \pm 6.6e-07$
n_1	-	$[0.0e+00, 1.0e+00]$	$7.83e-01 \pm 9.7e-03$	$6.87e-01 \pm 2.0e-02$	$7.04e-01 \pm 1.8e-02$
χ^2	-	-	$1.515e-02$	$1.513e-03$	$9.282e-05$

Analysis Results: Vasco_LNV3_4

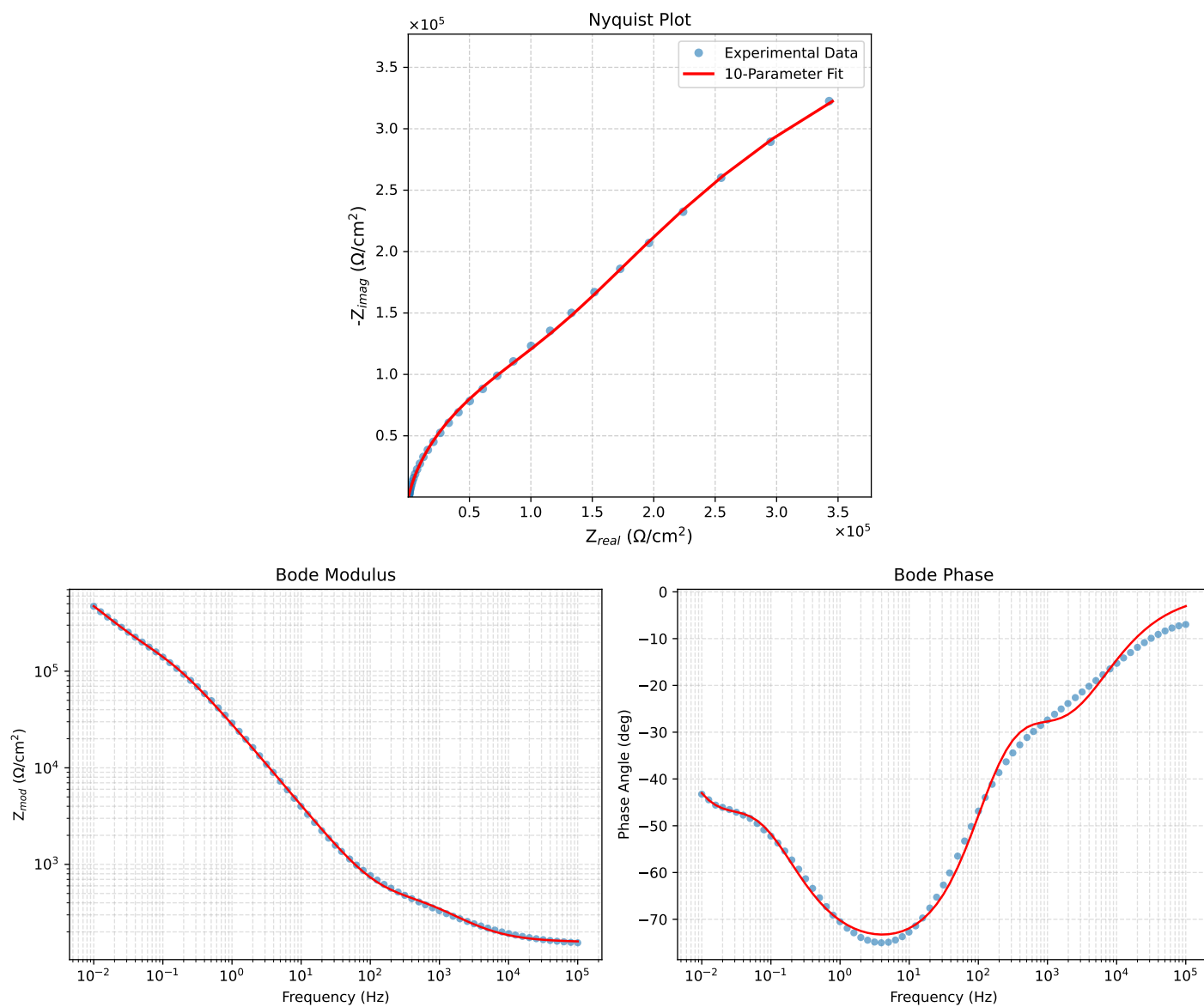


Figure 1: EIS Fit Results for Vasco_LNV3_4

Fitted Data: Vasco_LNV3_4

Frequency (Hz)	$\mathbf{z}_{real}$ (Ω)	$-\mathbf{z}_{imag}$ (Ω)	$\mathbf{z}_{mod}$ (Ω)	$\mathbf{z}_{phase}$ ($^\circ$)
1.0008e+05	1.8458e+02	3.0181e+00	1.8460e+02	-0.9368
7.9453e+04	1.8479e+02	3.6157e+00	1.8483e+02	-1.1210
6.3141e+04	1.8504e+02	4.3284e+00	1.8509e+02	-1.3400
5.0203e+04	1.8534e+02	5.1795e+00	1.8542e+02	-1.6007
3.9891e+04	1.8571e+02	6.2010e+00	1.8581e+02	-1.9125
3.1641e+04	1.8615e+02	7.4342e+00	1.8629e+02	-2.2870
2.5172e+04	1.8666e+02	8.8919e+00	1.8687e+02	-2.7273
2.0016e+04	1.8728e+02	1.0640e+01	1.8759e+02	-3.2515
1.5891e+04	1.8803e+02	1.2746e+01	1.8846e+02	-3.8781
1.2609e+04	1.8893e+02	1.5277e+01	1.8955e+02	-4.6228
1.0078e+04	1.8997e+02	1.8206e+01	1.9084e+02	-5.4742
8.0156e+03	1.9124e+02	2.1780e+01	1.9247e+02	-6.4973
6.3281e+03	1.9281e+02	2.6207e+01	1.9458e+02	-7.7403
5.0156e+03	1.9467e+02	3.1437e+01	1.9719e+02	-9.1737
3.9844e+03	1.9687e+02	3.7645e+01	2.0044e+02	-10.8251
3.1710e+03	1.9949e+02	4.5013e+01	2.0450e+02	-12.7153
2.5276e+03	2.0259e+02	5.3756e+01	2.0960e+02	-14.8604
1.9761e+03	2.0665e+02	6.5179e+01	2.1669e+02	-17.5055
1.5775e+03	2.1112e+02	7.7748e+01	2.2498e+02	-20.2172
1.2656e+03	2.1632e+02	9.2379e+01	2.3522e+02	-23.1253
9.9826e+02	2.2302e+02	1.1124e+02	2.4922e+02	-26.5089
7.9688e+02	2.3065e+02	1.3269e+02	2.6609e+02	-29.9118
6.2779e+02	2.4033e+02	1.5992e+02	2.8867e+02	-33.6408
5.0551e+02	2.5084e+02	1.8947e+02	3.1436e+02	-37.0651
3.9800e+02	2.6472e+02	2.2847e+02	3.4968e+02	-40.7956
3.1550e+02	2.8094e+02	2.7401e+02	3.9244e+02	-44.2843
2.5240e+02	2.9957e+02	3.2628e+02	4.4295e+02	-47.4444
1.9862e+02	3.2356e+02	3.9357e+02	5.0950e+02	-50.5757
1.5836e+02	3.5080e+02	4.6988e+02	5.8639e+02	-53.2561
1.2556e+02	3.8423e+02	5.6344e+02	6.8198e+02	-55.7084
1.0045e+02	4.2267e+02	6.7088e+02	7.9293e+02	-57.7880
7.9003e+01	4.7234e+02	8.0948e+02	9.3721e+02	-59.7359
6.3345e+01	5.2713e+02	9.6208e+02	1.0970e+03	-61.2813
5.0223e+01	5.9600e+02	1.1535e+03	1.2983e+03	-62.6746
3.8422e+01	6.9291e+02	1.4220e+03	1.5818e+03	-64.0210
3.1250e+01	7.8309e+02	1.6710e+03	1.8454e+03	-64.8912
2.4934e+01	9.0017e+02	1.9932e+03	2.1871e+03	-65.6953
1.9862e+01	1.0414e+03	2.3802e+03	2.5981e+03	-66.3686
1.5625e+01	1.2212e+03	2.8700e+03	3.1190e+03	-66.9493
1.2401e+01	1.4305e+03	3.4363e+03	3.7222e+03	-67.3982
9.9311e+00	1.6720e+03	4.0849e+03	4.4138e+03	-67.7394
7.9449e+00	1.9629e+03	4.8589e+03	5.2404e+03	-68.0027
6.3174e+00	2.3224e+03	5.8056e+03	6.2529e+03	-68.1974
5.0080e+00	2.7631e+03	6.9512e+03	7.4802e+03	-68.3223
3.9457e+00	3.3133e+03	8.3593e+03	8.9919e+03	-68.3786
3.1587e+00	3.9363e+03	9.9252e+03	1.0677e+04	-68.3671
2.5040e+00	4.7254e+03	1.1868e+04	1.2774e+04	-68.2888
1.9981e+00	5.6590e+03	1.4110e+04	1.5202e+04	-68.1458
1.5847e+00	6.8294e+03	1.6841e+04	1.8173e+04	-67.9261
1.2669e+00	8.2112e+03	1.9960e+04	2.1583e+04	-67.6380
9.9904e-01	1.0015e+04	2.3874e+04	2.5889e+04	-67.2424
7.9234e-01	1.2193e+04	2.8389e+04	3.0896e+04	-66.7565
6.3345e-01	1.4787e+04	3.3497e+04	3.6616e+04	-66.1820
5.0403e-01	1.8056e+04	3.9575e+04	4.3499e+04	-65.4749
4.0064e-01	2.2127e+04	4.6657e+04	5.1638e+04	-64.6271
3.1672e-01	2.7312e+04	5.5019e+04	6.1425e+04	-63.5997
2.5202e-01	3.3570e+04	6.4298e+04	7.2534e+04	-62.4310
2.0032e-01	4.1324e+04	7.4786e+04	8.5444e+04	-61.0765
1.5890e-01	5.0919e+04	8.6517e+04	1.0039e+05	-59.5214
1.2601e-01	6.2571e+04	9.9295e+04	1.1737e+05	-57.7826
1.0016e-01	7.6319e+04	1.1276e+05	1.3616e+05	-55.9080
7.9449e-02	9.2484e+04	1.2690e+05	1.5702e+05	-53.9151
6.3174e-02	1.1062e+05	1.4121e+05	1.7938e+05	-51.9265
5.0134e-02	1.3074e+05	1.5591e+05	2.0347e+05	-50.0191
3.9826e-02	1.5217e+05	1.7108e+05	2.2896e+05	-48.3487
3.1630e-02	1.7475e+05	1.8752e+05	2.5632e+05	-47.0183
2.5121e-02	1.9866e+05	2.0628e+05	2.8639e+05	-46.0781
1.9955e-02	2.2483e+05	2.2850e+05	3.2057e+05	-45.4634
1.5852e-02	2.5508e+05	2.5489e+05	3.6060e+05	-44.9795
1.2591e-02	2.9185e+05	2.8523e+05	4.0808e+05	-44.3431
1.0001e-02	3.3761e+05	3.1780e+05	4.6366e+05	-43.2694

Analysis Results: Vasco_LNV3_5

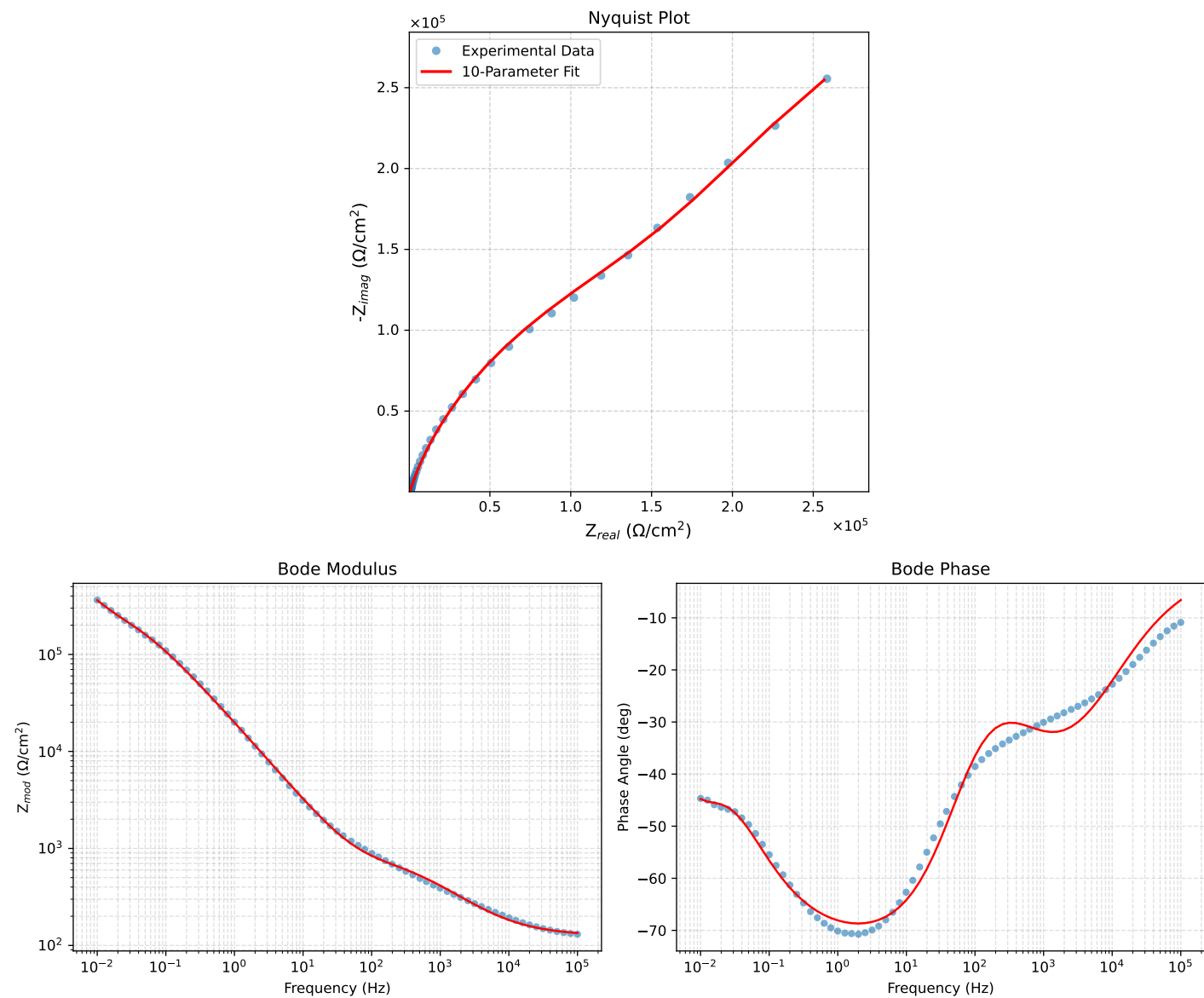


Figure 2: EIS Fit Results for Vasco_LNV3_5

Fitted Data: Vasco_LNV3_5

Frequency (Hz)	$\mathbf{z}_{real}$ (Ω)	$-\mathbf{z}_{imag}$ (Ω)	$\mathbf{z}_{mod}$ (Ω)	$\mathbf{z}_{phase}$ ($^{\circ}$)
1.0008e+05	1.3338e+02	1.5402e+01	1.3427e+02	-6.5870
7.9453e+04	1.3490e+02	1.7982e+01	1.3609e+02	-7.5927
6.3141e+04	1.3669e+02	2.0967e+01	1.3829e+02	-8.7209
5.0203e+04	1.3880e+02	2.4422e+01	1.4093e+02	-9.9793
3.9891e+04	1.4129e+02	2.8432e+01	1.4412e+02	-11.3776
3.1641e+04	1.4427e+02	3.3105e+01	1.4802e+02	-12.9235
2.5172e+04	1.4776e+02	3.8423e+01	1.5267e+02	-14.5764
2.0016e+04	1.5190e+02	4.4545e+01	1.5830e+02	-16.3442
1.5891e+04	1.5685e+02	5.1605e+01	1.6512e+02	-18.2120
1.2609e+04	1.6274e+02	5.9677e+01	1.7334e+02	-20.1383
1.0078e+04	1.6951e+02	6.8538e+01	1.8284e+02	-22.0151
8.0156e+03	1.7770e+02	7.8728e+01	1.9436e+02	-23.8949
6.3281e+03	1.8775e+02	9.0515e+01	2.0843e+02	-25.7382
5.0156e+03	1.9949e+02	1.0339e+02	2.2469e+02	-27.3962
3.9844e+03	2.1320e+02	1.1737e+02	2.4338e+02	-28.8340
3.1710e+03	2.2915e+02	1.3238e+02	2.6464e+02	-30.0160
2.5276e+03	2.4755e+02	1.4826e+02	2.8855e+02	-30.9179
1.9761e+03	2.7065e+02	1.6632e+02	3.1767e+02	-31.5715
1.5775e+03	2.9477e+02	1.8330e+02	3.4711e+02	-31.8746
1.2656e+03	3.2107e+02	1.9999e+02	3.7826e+02	-31.9180
9.9826e+02	3.5217e+02	2.1774e+02	4.1404e+02	-31.7277
7.9688e+02	3.8391e+02	2.3416e+02	4.4969e+02	-31.3809
6.2779e+02	4.1916e+02	2.5111e+02	4.8862e+02	-30.9252
5.0551e+02	4.5192e+02	2.6644e+02	5.2461e+02	-30.5231
3.9800e+02	4.8803e+02	2.8411e+02	5.6470e+02	-30.2061
3.1550e+02	5.2231e+02	3.0331e+02	6.0399e+02	-30.1438
2.5240e+02	5.5404e+02	3.2528e+02	6.4247e+02	-30.4180
1.9862e+02	5.8663e+02	3.5496e+02	6.8566e+02	-31.1775
1.5836e+02	6.1626e+02	3.9121e+02	7.2995e+02	-32.4079
1.2556e+02	6.4606e+02	4.3937e+02	7.8131e+02	-34.2184
1.0045e+02	6.7506e+02	4.9898e+02	8.3946e+02	-36.4703
7.9003e+01	7.0797e+02	5.8118e+02	9.1596e+02	-39.3831
6.3345e+01	7.4122e+02	6.7682e+02	1.0037e+03	-42.3997
5.0223e+01	7.8100e+02	8.0218e+02	1.1196e+03	-45.7667
3.8422e+01	8.3576e+02	9.8502e+02	1.2918e+03	-49.6864
3.1250e+01	8.8674e+02	1.1595e+03	1.4597e+03	-52.5931
2.4934e+01	9.5375e+02	1.3900e+03	1.6858e+03	-55.5444
1.9862e+01	1.0362e+03	1.6719e+03	1.9669e+03	-58.2091
1.5625e+01	1.1437e+03	2.0338e+03	2.3334e+03	-60.6484
1.2401e+01	1.2721e+03	2.4574e+03	2.7671e+03	-62.6312
9.9311e+00	1.4239e+03	2.9472e+03	3.2731e+03	-64.2132
7.9449e+00	1.6109e+03	3.5363e+03	3.8859e+03	-65.5091
6.3174e+00	1.8473e+03	4.2618e+03	4.6449e+03	-66.5651
5.0080e+00	2.1433e+03	5.1448e+03	5.5734e+03	-67.3835
3.9457e+00	2.5203e+03	6.2357e+03	6.7258e+03	-67.9927
3.1587e+00	2.9550e+03	7.4542e+03	8.0185e+03	-68.3757
2.5040e+00	3.5146e+03	8.9711e+03	9.6350e+03	-68.6061
1.9981e+00	4.1867e+03	1.0728e+04	1.1516e+04	-68.6803
1.5847e+00	5.0408e+03	1.2873e+04	1.3824e+04	-68.6151
1.2669e+00	6.0613e+03	1.5328e+04	1.6483e+04	-68.4236
9.9904e-01	7.4077e+03	1.8414e+04	1.9848e+04	-68.0861
7.9234e-01	9.0493e+03	2.1981e+04	2.3771e+04	-67.6232
6.3345e-01	1.1020e+04	2.6021e+04	2.8258e+04	-67.0479
5.0403e-01	1.3519e+04	3.0832e+04	3.3665e+04	-66.3239
4.0064e-01	1.6648e+04	3.6444e+04	4.0067e+04	-65.4490
3.1672e-01	2.0650e+04	4.3079e+04	4.7772e+04	-64.3892
2.5202e-01	2.5496e+04	5.0451e+04	5.6527e+04	-63.1893
2.0032e-01	3.1518e+04	5.8798e+04	6.6712e+04	-61.8072
1.5890e-01	3.8987e+04	6.8158e+04	7.8521e+04	-60.2300
1.2601e-01	4.8085e+04	7.8389e+04	9.1962e+04	-58.4741
1.0016e-01	5.8859e+04	8.9212e+04	1.0688e+05	-56.5844
7.9449e-02	7.1590e+04	1.0063e+05	1.2350e+05	-54.5707
6.3174e-02	8.5957e+04	1.1221e+05	1.4135e+05	-52.5464
5.0134e-02	1.0199e+05	1.2407e+05	1.6061e+05	-50.5785
3.9826e-02	1.1913e+05	1.3618e+05	1.8094e+05	-48.8218
3.1630e-02	1.3712e+05	1.4909e+05	2.0255e+05	-47.3950
2.5121e-02	1.5585e+05	1.6360e+05	2.2595e+05	-46.3894
1.9955e-02	1.7574e+05	1.8075e+05	2.5211e+05	-45.8054
1.5852e-02	1.9793e+05	2.0152e+05	2.8246e+05	-45.5156
1.2591e-02	2.2421e+05	2.2637e+05	3.1861e+05	-45.2749
1.0001e-02	2.5674e+05	2.5479e+05	3.6171e+05	-44.7815

Analysis Results: Vasco_LNV3_7d_

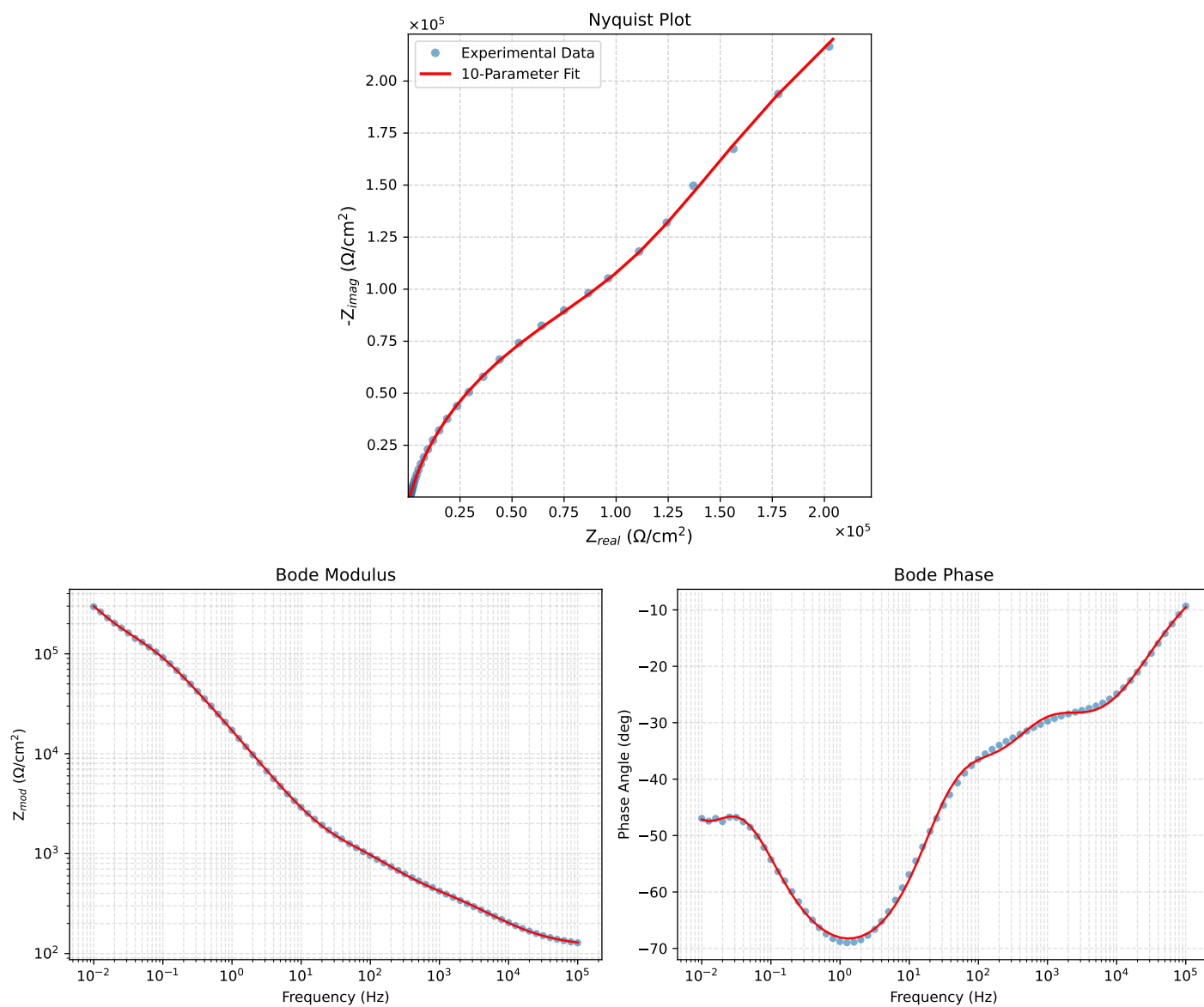


Figure 3: EIS Fit Results for Vasco_LNV3_7d_

Fitted Data: Vasco_LNV3_7d_

Frequency (Hz)	$\mathbf{z}_{real}$ (Ω)	$-\mathbf{z}_{imag}$ (Ω)	$\mathbf{z}_{mod}$ (Ω)	$\mathbf{z}_{phase}$ ($^{\circ}$)
1.0002e+05	1.2716e+02	2.1342e+01	1.2894e+02	-9.5273
7.9512e+04	1.2939e+02	2.4858e+01	1.3175e+02	-10.8752
6.3105e+04	1.3207e+02	2.8936e+01	1.3520e+02	-12.3582
5.0215e+04	1.3523e+02	3.3558e+01	1.3934e+02	-13.9360
3.9902e+04	1.3904e+02	3.8858e+01	1.4437e+02	-15.6141
3.1699e+04	1.4360e+02	4.4873e+01	1.5044e+02	-17.3537
2.5137e+04	1.4908e+02	5.1697e+01	1.5779e+02	-19.1250
1.9980e+04	1.5555e+02	5.9229e+01	1.6644e+02	-20.8455
1.5879e+04	1.6323e+02	6.7543e+01	1.7665e+02	-22.4791
1.2598e+04	1.7236e+02	7.6666e+01	1.8865e+02	-23.9790
1.0020e+04	1.8294e+02	8.6351e+01	2.0230e+02	-25.2681
7.9282e+03	1.9548e+02	9.6807e+01	2.1813e+02	-26.3462
6.3079e+03	2.0948e+02	1.0740e+02	2.3540e+02	-27.1445
5.0347e+03	2.2498e+02	1.1805e+02	2.5407e+02	-27.6869
3.9931e+03	2.4256e+02	1.2907e+02	2.7476e+02	-28.0173
3.1441e+03	2.6223e+02	1.4043e+02	2.9746e+02	-28.1703
2.5195e+03	2.8156e+02	1.5103e+02	3.1951e+02	-28.2095
2.0008e+03	3.0253e+02	1.6240e+02	3.4336e+02	-28.2269
1.5807e+03	3.2462e+02	1.7485e+02	3.6871e+02	-28.3078
1.2536e+03	3.4692e+02	1.8858e+02	3.9486e+02	-28.5278
1.0007e+03	3.6929e+02	2.0409e+02	4.2193e+02	-28.9278
7.9003e+02	3.9392e+02	2.2347e+02	4.5289e+02	-29.5667
6.2881e+02	4.1951e+02	2.4593e+02	4.8628e+02	-30.3800
5.0223e+02	4.4731e+02	2.7216e+02	5.2360e+02	-31.3180
4.0015e+02	4.7899e+02	3.0312e+02	5.6685e+02	-32.3265
3.1550e+02	5.1700e+02	3.4026e+02	6.1893e+02	-33.3504
2.5202e+02	5.5827e+02	3.7957e+02	6.7508e+02	-34.2115
2.0032e+02	6.0613e+02	4.2348e+02	7.3941e+02	-34.9404
1.5801e+02	6.6136e+02	4.7246e+02	8.1278e+02	-35.5410
1.2556e+02	7.1948e+02	5.2349e+02	8.977e+02	-36.0396
9.9734e+01	7.8074e+02	5.7916e+02	9.7211e+02	-36.5685
7.9449e+01	8.4264e+02	6.4085e+02	1.0586e+03	-37.2540
6.2919e+01	9.0623e+02	7.1476e+02	1.1542e+03	-38.2635
4.9867e+01	9.6915e+02	8.0419e+02	1.2594e+03	-39.6854
3.8422e+01	1.0398e+03	9.3080e+02	1.3955e+03	-41.8343
3.1250e+01	1.0972e+03	1.0571e+03	1.5236e+03	-43.9336
2.4934e+01	1.1637e+03	1.2283e+03	1.6920e+03	-46.5481
2.0032e+01	1.2345e+03	1.4344e+03	1.8925e+03	-49.2847
1.5625e+01	1.3267e+03	1.7266e+03	2.1774e+03	-52.4623
1.2467e+01	1.4258e+03	2.0566e+03	2.5025e+03	-55.2669
9.9734e+00	1.5431e+03	2.4550e+03	2.8997e+03	-57.8480
7.9719e+00	1.6860e+03	2.9411e+03	3.3901e+03	-60.1755
6.3516e+00	1.8638e+03	3.5389e+03	3.9997e+03	-62.2264
5.0134e+00	2.0935e+03	4.2954e+03	4.7784e+03	-64.0161
3.9860e+00	2.3713e+03	5.1843e+03	5.7009e+03	-65.4205
3.1758e+00	2.7145e+03	6.2453e+03	6.8097e+03	-66.5082
2.5161e+00	3.1550e+03	7.5538e+03	8.1862e+03	-67.3311
1.9955e+00	3.7073e+03	9.1203e+03	9.8450e+03	-67.8787
1.5879e+00	4.3935e+03	1.0969e+04	1.1816e+04	-68.1713
1.2581e+00	5.2781e+03	1.3217e+04	1.4232e+04	-68.2307
9.9777e-01	6.3971e+03	1.5881e+04	1.7121e+04	-68.0590
7.9274e-01	7.8095e+03	1.9007e+04	2.0549e+04	-67.6638
6.2903e-01	9.6188e+03	2.2703e+04	2.4657e+04	-67.0390
4.9888e-01	1.1931e+04	2.7023e+04	2.9540e+04	-66.1788
3.9860e-01	1.4769e+04	3.1841e+04	3.5100e+04	-65.1160
3.1672e-01	1.8439e+04	3.7451e+04	4.1745e+04	-63.7867
2.5148e-01	2.3062e+04	4.3755e+04	4.9461e+04	-62.2079
1.9998e-01	2.8751e+04	5.0625e+04	5.8219e+04	-60.4071
1.5879e-01	3.5716e+04	5.8032e+04	6.8142e+04	-58.3897
1.2601e-01	4.4009e+04	6.5782e+04	7.9146e+04	-56.2168
1.0016e-01	5.3486e+04	7.3617e+04	9.0996e+04	-53.9997
7.9503e-02	6.4076e+04	8.1525e+04	1.0369e+05	-51.8335
6.3140e-02	7.5388e+04	8.9484e+04	1.1701e+05	-49.8867
5.0123e-02	8.7135e+04	9.7818e+04	1.3100e+05	-48.3057
3.9833e-02	9.9030e+04	1.0703e+05	1.4581e+05	-47.2222
3.1638e-02	1.1121e+05	1.1793e+05	1.6209e+05	-46.6794
2.5126e-02	1.2408e+05	1.3135e+05	1.8069e+05	-46.6308
1.9957e-02	1.3844e+05	1.4802e+05	2.0267e+05	-46.9152
1.5849e-02	1.5553e+05	1.6845e+05	2.2927e+05	-47.2837
1.2590e-02	1.7685e+05	1.9267e+05	2.6152e+05	-47.4514
1.0001e-02	2.0413e+05	2.2013e+05	3.0022e+05	-47.1599